ANDREW TARRH

(781) 602-0523 · atarrh@gmail.com

atarrh.io · linkedin.com/in/andrewtarrh · github.com/atarrh

OBJECTIVE

Full-stack engineer with a background in petabyte-scale data infrastructure at Amazon, now doing contract work focused on infrastructure tooling, workflow automation, and feature delivery across React, Node, Python, and SQL. Spent five years building the data platform behind Alexa's ML systems, then shifted toward full-stack engineering—shipping features, hardening pipelines, and building internal tools across civic tech and higher education. Gravitating toward mission-driven roles at the intersection of systems thinking and product delivery.

PROFESSIONAL EXPERIENCE

Full Stack Engineer (Infrastructure) April 2026 – Present
Arizona State University (via Dexian) *Remote*

- Hardening Salesforce enrollment sync pipeline—fixing race conditions, deduplication, and retry failures across Salesforce, Canvas, and BigQuery.
- Improving record consistency between upstream and downstream data sources from unknown parity to 99.7%+, enabling reliable cross-system reporting.
- Replacing hardcoded permissions management with a web-forward admin UI, eliminating manual configuration and enabling non-technical staff to manage access directly.

Software Engineer November 2025 – April 2026
WeVote *Remote*

- Developed voter-facing UI components in a Node/React/Flux application and shipped bug fixes and backend improvements to Python/Django API endpoints.

Independent Contractor 2021 – 2025
Mercor, Interview Kickstart *Remote*

- Mercor: Evaluated and improved LLM performance for code generation through expert feedback on programming prompts.
- Interview Kickstart: Mentored mid-career professionals on scalable infrastructure, ETL, and data engineering.

Software Development Engineer / SDE II September 2016 – September 2021
Amazon — Alexa AI *Cambridge, MA*

- Designed and operated petabyte-scale data infrastructure across S3, Athena, Redshift, Airflow, and Elasticsearch for large-scale ML training and data discovery.
- Led migration from monolithic Redshift warehouses to serverless data lake architecture, improving privacy controls and reducing costs by $50K+/year.
- Built a secure cross-region data replication service with encryption-in-transit for globally distributed model training.
- Automated ML pipeline deployment using microservices and testing frameworks, reducing NLU model deployment failures by 75%.
- Served as technical lead, security certifier, and operational excellence liaison while mentoring engineers across the organization.

EDUCATION

Boston University May 2016
M.S. Computer Science — B.A. Computer Science & Astronomy

SKILLS, TOOLS & TECHNOLOGIES

Languages	JavaScript/TypeScript, Python, Java, SQL, Bash, HTML/CSS
Web & APIs	React, Node.js, NestJS/Express, Django, Flask, Vue.js
Data & Infrastructure	PostgreSQL, S3, DynamoDB, Kinesis, Redshift, Athena, Elasticsearch, Airflow, Spark
Cloud & DevOps	AWS (Lambda, EKS/ECS, Fargate, CloudWatch), Linux, CI/CD
Security & Compliance	KMS, IAM, VPC, FERPA, CCPA, GDPR, Security Audits